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To be cited as: *Angew. Chem. Int. Ed.* **2025**, e202505155

Link to VoR: https://doi.org/10.1002/anie.202505155

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Deconstructive Functionalization of Cyclic Ketones via Electrochemical Interrupted Dowd-Beckwith Reaction

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Abstract: Deconstructive functionalization of cyclic molecules has recently emerged as a prominent strategy to access unique architectures that are challenging to prepare through traditional methods. While significant progress has been made in the deconstructive functionalization of cyclic alcohols and amines, the strategies for deconstructing cyclic ketones remain largely unexplored. The Dowd-Beckwith reaction, a ring-expansion of cyclic ketones, is a powerful method for synthesizing medium- and large-ring compounds. Herein, we developed the first interrupted Dowd-Beckwith (IDB) reaction for highly regioselective deconstructive functionalization of cyclic ketones using electricity as the sole oxidant. This protocol is widely applicable for the deconstruction of small, medium-sized, and macrocyclic ketones bearing a diverse range of functional groups. Remarkably, various naturally occurring complex cyclic ketones were successfully deconstructed into acyclic molecules, which are difficult to access by known strategies. The method was applied to the asymmetric synthesis of planococcol, citrilol acetate, maconelliol, and its derivatives. Furthermore, the functional groups incorporated during the transformation provided versatile handles for subsequent synthetic modifications. Mechanistic experiments and computational studies support an oxidative radical-polar crossover followed by deconstructive functionalization.

Introduction

Skeletal remodelling, an approach that alters the core framework of a molecule to attain skeletal diversity, has garnered significant attention in medicinal chemistry discovery programmes, complementing strategies that are focused on peripheral functionalizations.[1] This strategy forms the foundation of recent drug discovery efforts aimed at developing structurally distinct scaffolds with augmented physiochemical properties and pharamacodynamics.[1a, b, 2] Among skeletal remodelling, the deconstructive functionalization of saturated carbocycles and heterocycles is an efficient strategy, which involves the cleavage of carbon–carbon (C_{sp3}–C_{sp3}) bonds, followed by new bond forming events at one or more constituent carbons, giving access to synthetically unique architectures that are challenging to prepare through traditional methods (Figure 1A).[3] Cyclic ketones not only serve as versatile building blocks in chemical synthesis but are also found in agrochemicals, pharmaceuticals, and complex natural products.[4] Historically, the chemical diversification of these motifs has expanded the synthetic arsenal, yielding a wealth of valuable derivatives that span applications from therapeutic agents to biologically active compounds.[4] Despite their importance, the diversification of unstrained cyclic ketones through deconstructive functionalization strategies remains largely unexplored,[5] in contrast to the well-established methods available for cyclic alcohols[6] and -amines[7] (Figure 1A). Consequently, innovative strategies that use sustainable techniques for the deconstructive functionalization of cyclic ketones are highly sought after since they have the potential to unlock new chemical space.

Owing to their operational simplicity and broad functional group tolerance, radical-based strategies have emerged as powerful synthetic tools for constructing structurally diverse scaffolds with high efficiency and selectivity.[8] Despite the widespread occurrence and synthetic utility of ketones, the radical fragmentation of their α-carbon−carbon bonds remains a significant challenge due to the high thermodynamic and kinetic stability of the carbon−carbon single bond (BDE = 83 kcal/mol for cyclohexanone).[9] Traditionally, this cleavage has been achieved via UV-light irradiation in the classic Norrish-type I photochemical reaction.[10] Modern approaches rely on a controlled sequence involving carbonyl pre-activation, homolytic scission, and radical capture.[11] Among these, aromatization-driven strategies have been particularly prominent, wherein the carbonyl group undergoes condensation to form a pre-aromatic heterocyclic intermediate, followed by a redox-triggered fragmentation.[12] However, these strategies often encounter atom economy issues and regioselective ring-opening issues when unsymmetrical ketones are employed.[11c, 12c, g] We envisioned a novel and mechanistically distinct deconstructive strategy that

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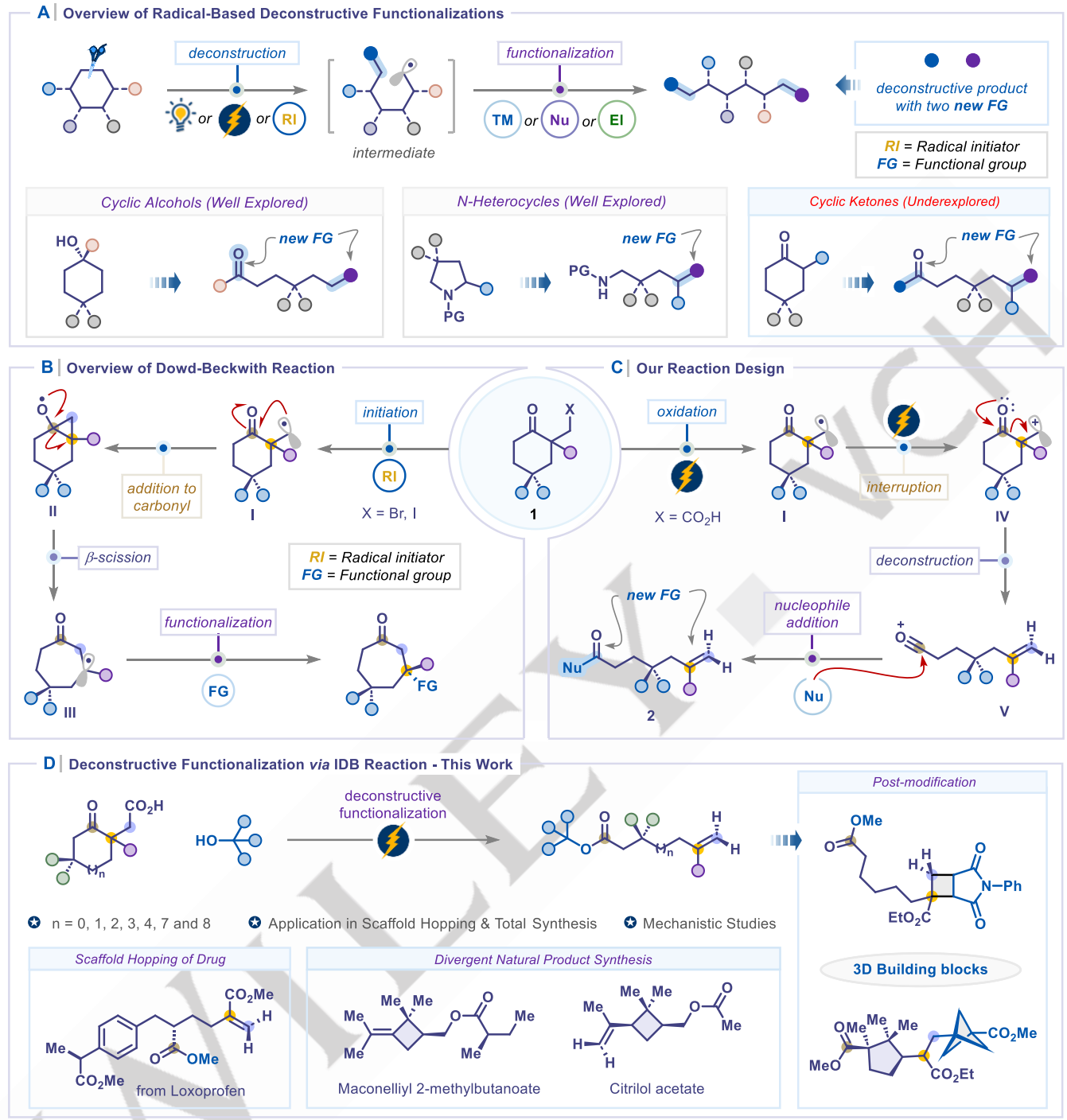


Figure 1: (A) Overview of radical-based deconstructive functionalizations. (B) Overview of Dowd-Beckwith reaction. (C) Our reaction design. (D) This work: Deconstructive functionalization *via* IDB reaction.

allows convenient and completely regioselective ring-opening *via* the interruption of the Dowd-Beckwith reaction, a powerful radical-based ring-expansion strategy that used for synthesizing medium to large-sized carbocyclic scaffolds (Figure 1B).[13]

Electro-organic synthesis has undergone a significant resurgence in recent years and evolved as a powerful tool for sustainable chemical synthesis, requiring only electrical energy as the primary input to provide fast access to complex, high value products where scale-up can be accomplished with a small footprint, avoids reagent waste generation and increases atom economy.[14] Within this field, radical-polar crossover (RPC) refers to a specific class of reactions involving the interlacing of radical and polar/ionic species, known as net-oxidation or net-reduction.[15] We anticipated that electrochemical oxidative radical-polar crossover could interrupt the Dowd-Beckwith reaction, enabling the regioselective deconstructive functionalization of cyclic ketones under ambient conditions. Mechanistically, interrupting the Dowd-Beckwith reaction entails the

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oxidation of radical intermediate **I**. We hypothesized that acid **1** would first undergo single-electron transfer (SET) oxidation under electrolysis, followed by decarboxylation to generate radical **I**. Subsequent oxidation of **I** to carbocation **IV** would facilitate the ring-opening of cyclic ketones to give acylium intermediate **V** that could be trapped with a nucleophile to give the acyclic product **2** with two versatile functional groups, olefin and carbonyl (Figure 1C). The key to the success of this hypothesis would be achieving the rapid oxidation of radical **I** before it forms cyclopropyloxy radical **II**. In defiance of this challenge, herein, we present our success in developing the first electrochemical strategy for the deconstructive functionalization of cyclic ketones *via* an interrupted Dowd-Beckwith reaction (Figure 1D). Our strategy is broadly applicable to a range of 5-, 6-, 7-, 8-, 9-, 12-, and 13-membered cyclic ketones, accommodating a diverse range of functional groups. Notably, several complex ketones were efficiently deconstructed into acyclic molecules. The synthetic utility of this methodology was demonstrated in the asymmetric synthesis of planococcol, citrilol acetate, maconelliol, and its derivatives. Control experiments and DFT studies suggest the radical-polar crossover mechanism *via* electrochemical net-oxidation.

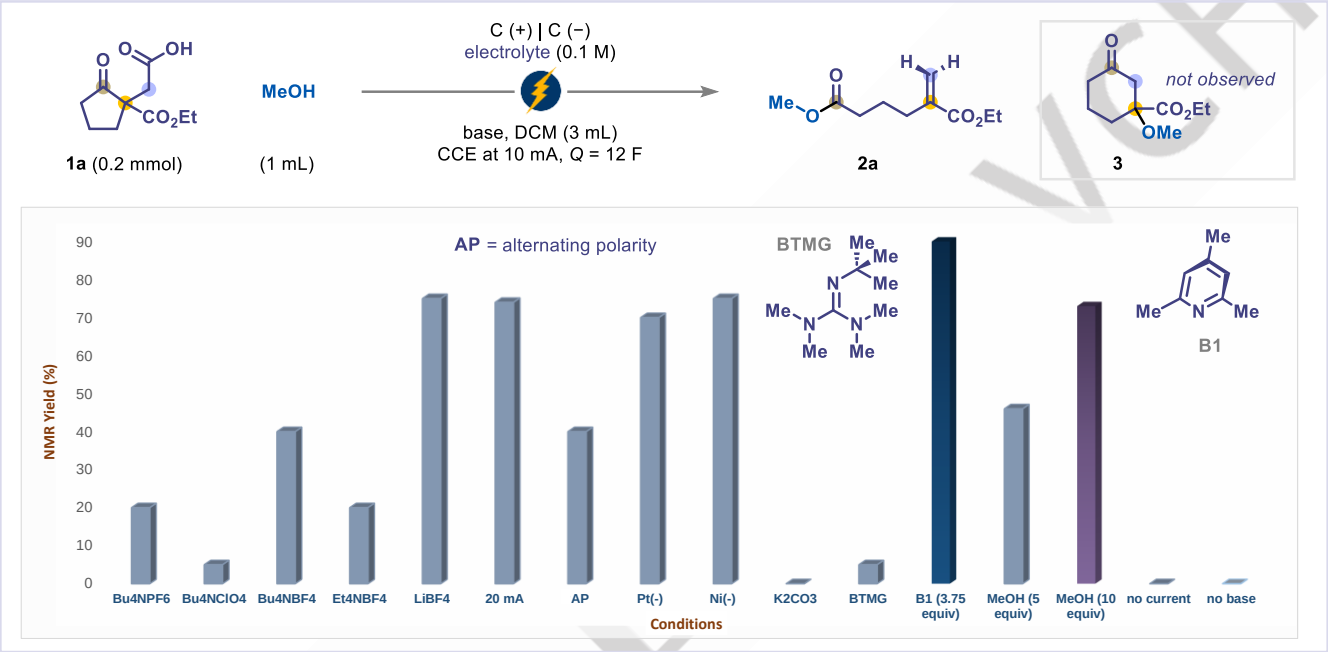


Figure 2: Optimization of the reaction conditions. Carboxylic acid **1a** (0.2 mmol), electrolyte (0.1 M), base (2.0 equiv), 3:1 DCM:MeOH (0.05M), Undivided cell, constant current electrolysis (CCE) at 10 mA, Q = 12 F. AP = Alternating polarity at interval of 5 seconds. Yields were determined by ^{1}H-NMR using CH_2Br_2 as the internal standard.

Results and Discussion

We began our investigation by electrolysis of the carboxylic acid **1a** in the presence of 2,4,6-collidine as base and $^{n}Bu_4NPF_6$ as an electrolyte in a 3:1 DCM/MeOH mixture at a constant current of 10 mA. Pleasingly, the desired deconstructive product **2a** was obtained in 20% yield (Figure 2, column 1). Among the electrolytes screened (Figure 2, columns 2-5), $LiBF_4$ emerged as the most effective, providing a yield of 75% (Figure 2, column 5). Performing the electrolysis with a 20-mA constant current gave a similar yield (Figure 2, column 6). The Baran group recently reported that alternating polarity (AP) offers precise control over reaction outcomes, enhancing yield,[16] and therefore, this was examined here, too. Carrying out electrolysis employing AP lowered the reaction yield (Figure 2, column 7). Switching the cathode material from graphite to platinum or Ni-foam yielded the desired product with similar efficiency (Figure 2, columns 8 and 9). However, Ni-foam was chosen for further screening due to its superior reproducibility.[17] Next, we evaluated various bases and their equivalents and found that 2,4,6-collidine afforded the desired product in 90% yield (Figure 2, column 12). Next, we screened various solvents using methanol as a reagent (see SI for complete optimization details). We were pleased to find that the desired product **2a** was obtained in 73% yield using 10 equiv of MeOH in DCM as the solvent (Figure 2, column14). This shows that MeOH is not required to be used as a solvent. Finally, control experiments underscored the essential role of both electricity and base in the reaction, as the absence of either component resulted in no reaction (Figure 2, columns 15 and 16).

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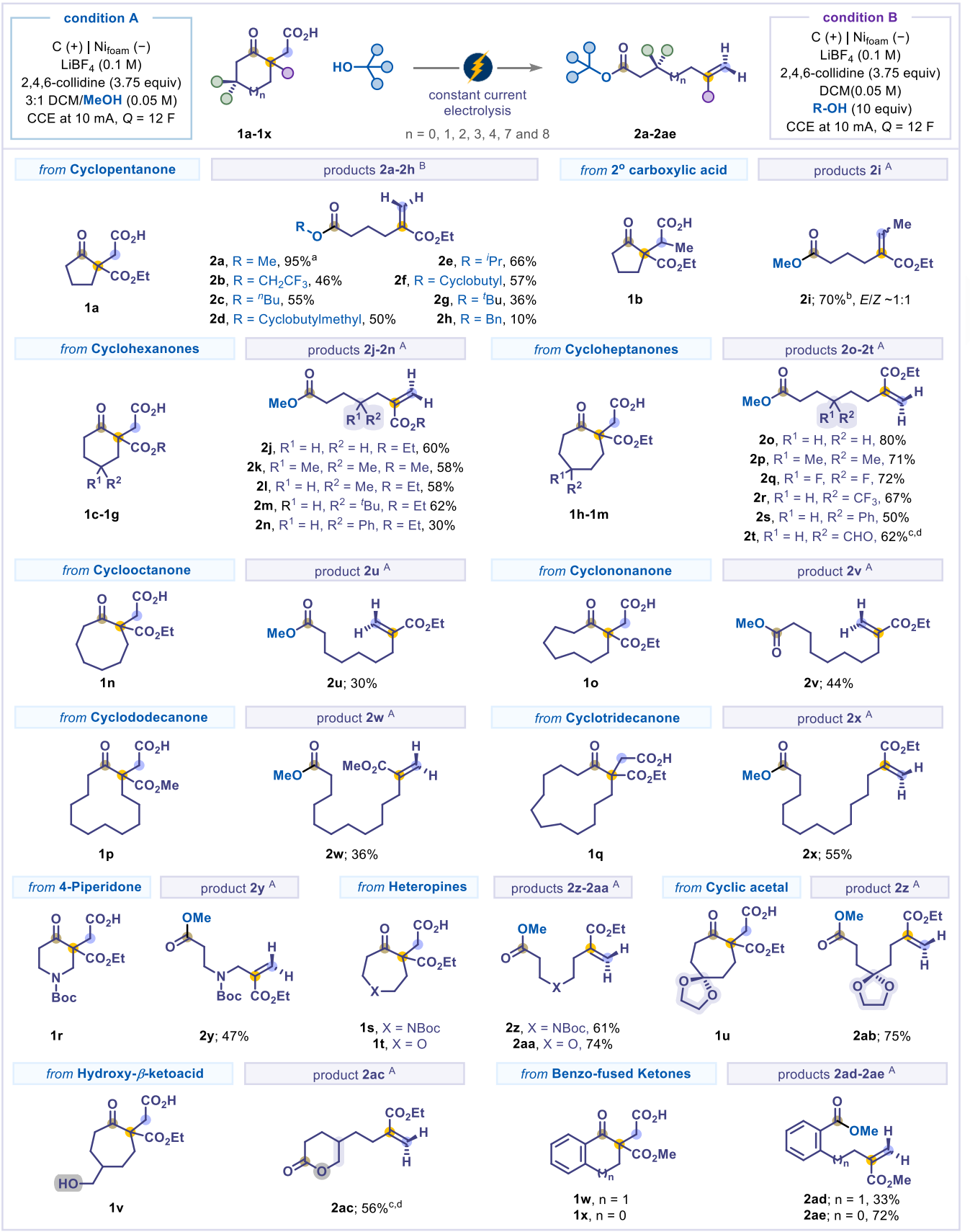


Figure 3: Substrate scope of alcohol nucleophiles and ketones. Condition A: carboxylic acid **1** (0.3 mmol), LiBF$_4$ (0.1 M), 2,4,6-collidine (3.75 equiv), 3:1 DCM:MeOH (0.05 M), Undivided cell: graphite (anode), Ni-foam (cathode), CCE at 10 mA, Q = 12 F. **Condition B:** carboxylic acid **1a** (0.3 mmol), alcohol (10 equiv), LiBF$_4$ (0.1 M), 2,4,6-collidine (3.75 equiv), DCM (0.05 M), Undivided cell: graphite (anode), Ni-foam (cathode), CCE at 10 mA, Q = 12 F. [a]Using condition A. [b]0.2 mmol scale. [c]0.15 mmol scale. [d]Reaction was performed in pure methanol. Yields are of isolated products.

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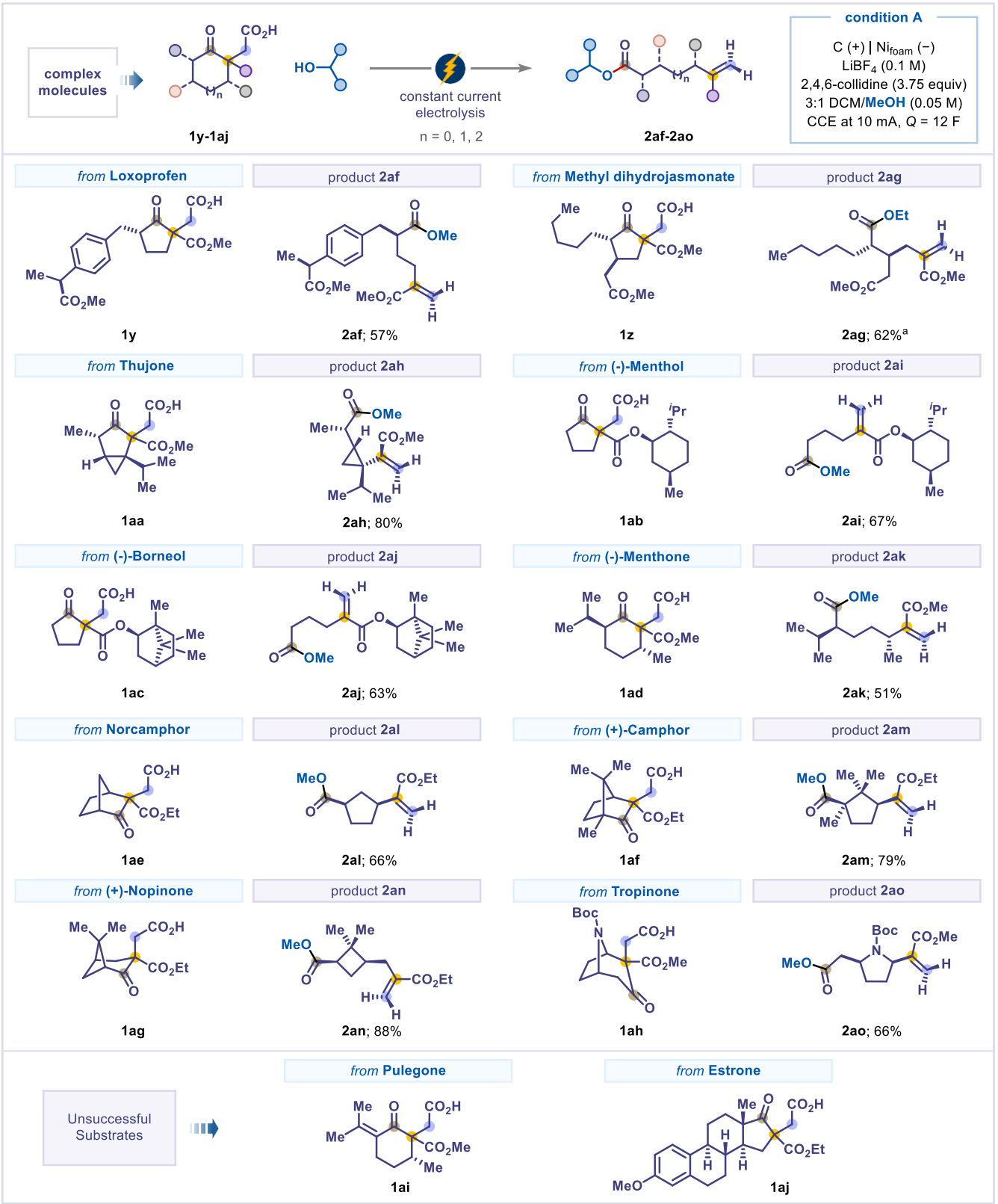


Figure 4: Substrate scope of complex cyclic ketones. Reaction condition: carboxylic acid **1** (0.3 mmol), LiBF$_4$ (0.1 M), 2,4,6-collidine (3.75 equiv), 3:1 DCM:MeOH (0.05 M), Undivided cell: graphite (anode), Ni-foam (cathode), CCE at 10 mA, Q = 12 F. Yields are of isolated products. [a]EtOH was used as a nucleophile.

Having the established optimal reaction conditions, we investigated the substrate scope of this deconstructive functionalization reaction using different alcohols as nucleophiles (Figure 3). *Primary* alcohols, including methanol, trifluoroethanol, *n*-butanol, and cyclobutylmethanol yielded the ring-opened products

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2a-2d in moderate to good yields. *Secondary* and *tertiary* alcohols were also amenable substrates under the reaction conditions (products **2e-2g**). The benzyl alcohol furnished the product **2h** in low yield due to its rapid oxidation to benzaldehyde. Unfortunately, amines and sulfur nucleophiles were unsuccessful in this reaction (See SI for more details). The *secondary* carboxylic acid **1b** successfully participated in the reaction to deliver the desired product **2i** in 70% yield with a 1:1 *E*/*Z* ratio. Subsequently, we evaluated the scope of various cyclic ketones. Cyclohexanones bearing dimethyl, *tert*-butyl, and phenyl substituents at C4 carbon delivered the corresponding deconstructive products **2j-2n** in good yields. Dimethyl, difluoro, trifluoromethyl, and phenyl-substituted cycloheptanones reacted well, affording the products **2o-2s** in good to excellent yields. A keto-acid-bearing aldehyde is well tolerated under oxidative conditions, resulting in the deconstructive product **2t** in 62% yield. Notably, medium-sized rings cyclooctanone and cyclononanone were also viable substrates (products **2u** and **2v**). We were pleased to find that macrocycles such as cyclododecanone and cyclotridecanone, which were not used previously in any deconstructive functionalization strategies, participated well in this transformation with moderate yields (products **2w** and **2x**). Ring-opening of the piperidone **1r** gave access to N-allylated *β*-amino acid derivative **2y**. Carboxylic acids **1s** and **1t** derived from azepan-4-one and oxepan-4-one also underwent the desired transformation successfully, furnishing the corresponding amino acid derivative **2z** and ether **2aa** in 61% and 74% yields, respectively. An acid-sensitive functional group such as cyclic acetal **1u** was well tolerated in this transformation to provide the product **2ab**. Interestingly, the OH-bearing substrate **1v** afforded the lactone **2ac** via intramolecular cyclization after deconstruction, along with a methanol addition product **2ac'** in 10% yield (See SI for the structure of **2ac'**). A benzo-fused bicyclic substrate, *α*-tetralone, was also a viable substrate (product **2ad**). Interestingly, the electrolysis of **1x** produced a 1:1 mixture of **2ae** and its double bond-reduced product **2ae'** (See SI for the structure of **2ae'**) in good yields. The formation of **2ae'** can be attributed to the net reduction of **2ae** at the cathode.

Cyclic ketones are omnipresent in natural products and pharmaceuticals. Next, we investigated substrates derived from naturally occurring cyclic ketones (Figure 4). Loxoprofen, a nonsteroidal anti-inflammatory drug (NSAID), successfully gave the ring-opening product **2af** in moderate yield, demonstrating a potential for late-stage functionalization. Electrolysis of methyl dihydrojasmonate yielded the deconstructive product **2ag** with *anti*-vicinal substituents. Thujone, a fused ketone, afforded the corresponding highly substituted cyclopropane **2ah** in excellent yield; such a substitution pattern on cyclopropane is challenging to achieve by traditional methods. Menthol and borneol pendant cyclopentanones **1ab** and **1ac** were also viable substrates, providing products **2ai** and **2aj** in 67% and 63% yields, respectively. The product **2ak,** which has stereocenters at C3 and C6, was obtained through the ring-opening of acid **1ad** derived from (−)-menthone. Bicyclic bridged ketones accessed from norcamphor, (+)-camphor, and (+)-nopinone were successfully electrolyzed to afford the *syn*-1,3-disubstituted cyclopentanes **2al** and **2am**, and highly substituted cyclobutane **2an** in excellent yields. Finally, the bicyclic tropinone-derived substrate **1ah** bearing *N*-Boc substituent provided the *syn*-2,5-disubstituted pyrrolidine **2ao** in good yield. Unfortunately, substrates **1ai** and **1aj** didn't give the desired products under the standard conditions.

Application

To illustrate the practicality of this deconstructive functionalization reaction, gram-scale reactions were performed with the standard conditions. Electrolysis of bicyclic ketone **1ag** on a 15 mmol scale provided the *syn*-1,3-disubstituted cyclobutane **2an** in excellent yield (3.3 g, 87%). Furthermore, **1ag** was electrolyzed on a 2 mmol scale in three batches, yielding similar results that highlighted the reproducibility of the methodology (Figure 5A). Next, we shifted our attention to demonstrating the synthetic applicability of the developed method. We established a unified divergent strategy for asymmetric syntheses of dimethylcyclobutane-containing natural products, planococcol, maconelliol, and their ester derivatives (Figure 5B).[18] These compounds serve as sex pheromones for mealybugs affecting various plants, such as vegetables, fruits, and ornamental crops. They are widely utilized in agriculture for detecting, monitoring, and implementing attract-and-kill strategies to manage pest populations.[18] Through retrosynthetic analysis, we anticipated that these natural products (**4-8**) could be synthesized from a common intermediate *syn*-1,3-disubstituted cyclobutane **2an** that was obtained from the electrolysis of **1ag** accessed from (1*R*)-(+)-nopinone (See SI). Our efforts to isomerize the terminal olefin in cyclobutane **2an** were ineffective using previously reported conditions with DBU or Co-salen/silane systems (Figure 5C).[19] Pleasingly, a catalytic amount of rhodium trichloride ($RhCl_3$) in ethanol yielded the olefin isomerize product **9** in excellent yield.[20] Subsequent ozonolysis and in situ reduction afforded the alcohol **10**. The addition of methyl Grignard gave the corresponding *tertiary* alcohol **12** in 95% yield without any racemization. Selective protection of *primary* alcohol, dehydration of *tertiary* alcohol with thionyl chloride, and in-situ deprotection using TBAF in a one-pot fashion yielded the natural product planococcol **4** in overall yield of 24% from commercially available (1*R*)-(+)-nopinone. The citrilol

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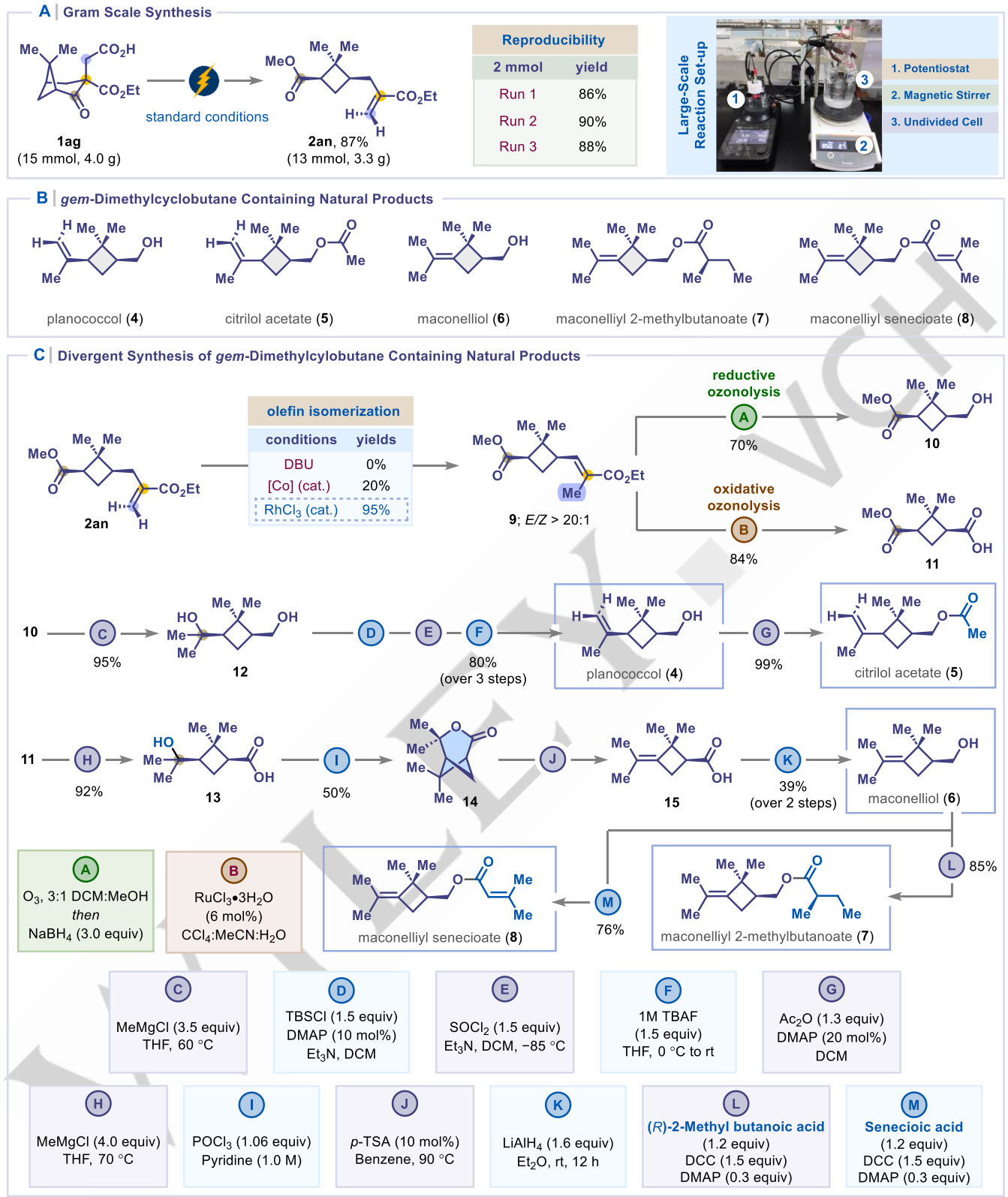


Figure 5: (A) Gram-scale synthesis. (B) Natural products containing *gem*-dimethylcyclobutane. (C) Total synthesis of divergent natural products.

acetate **5** was obtained from **4** in quantitative yield using acetic anhydride. We then turned our focus to synthesizing maconelliol **6** and its derivatives. Oxidative ozonolysis of **9** gave the carboxylic acid **11** in 84% yield. Treating ester **11** with methyl Grignard reagent afforded *tertiary* alcohol **13**, which was then converted into bicyclic lactone **14** using $POCl_3$ in pyridine.[18b] Acid-catalyzed ring-opening followed by reduction with $LiAlH_4$ gave maconelliol **6** in 8% overall yield from (1*R*)-(+)-nopinone. Finally, esterification of (*R*)-2-

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methylbutanoic acid and senecioic acid with maconelliol **6** yielded natural products maconelliyl-2-methylbutanoate **7** and maconelliyl senecioate **8** in 85% and 76% yields, respectively.

We further highlighted the synthetic potential of the obtained products by efficiently transforming them into useful building blocks (Figure 6). Cyclobutane-fused scaffold **16** was accessed in 90% yield by photochemical [2+2] cycloaddition of **2o** with *N*-phenyl maleimide using iridium-photosensitizer. Decarboxylative Giese addition of oleic acid to **2o** afforded the C_{25}-chain carboxyl acid derivative **17** in 64% yield. Structurally rigid motifs are gaining prominence as bioisosteres for planar molecules in medicinal chemistry and drug discovery since they enhance properties such as lipophilicity, solubility, and metabolic stability compared to flat molecular strcutures.[21] Photo-irradiation of bicyclo[1.1.1]pentane carboxylic acid with the vinylcyclopentane derivative **2am** in the presence of 4-CzIPN yielded highly functionalized cyclopentyl-bicyclo[1.1.1]pentane **18** in 71% yield. Finally, diol **19** was obtained by reduction of diester **2am** in good yield using $LiAlH_4$.

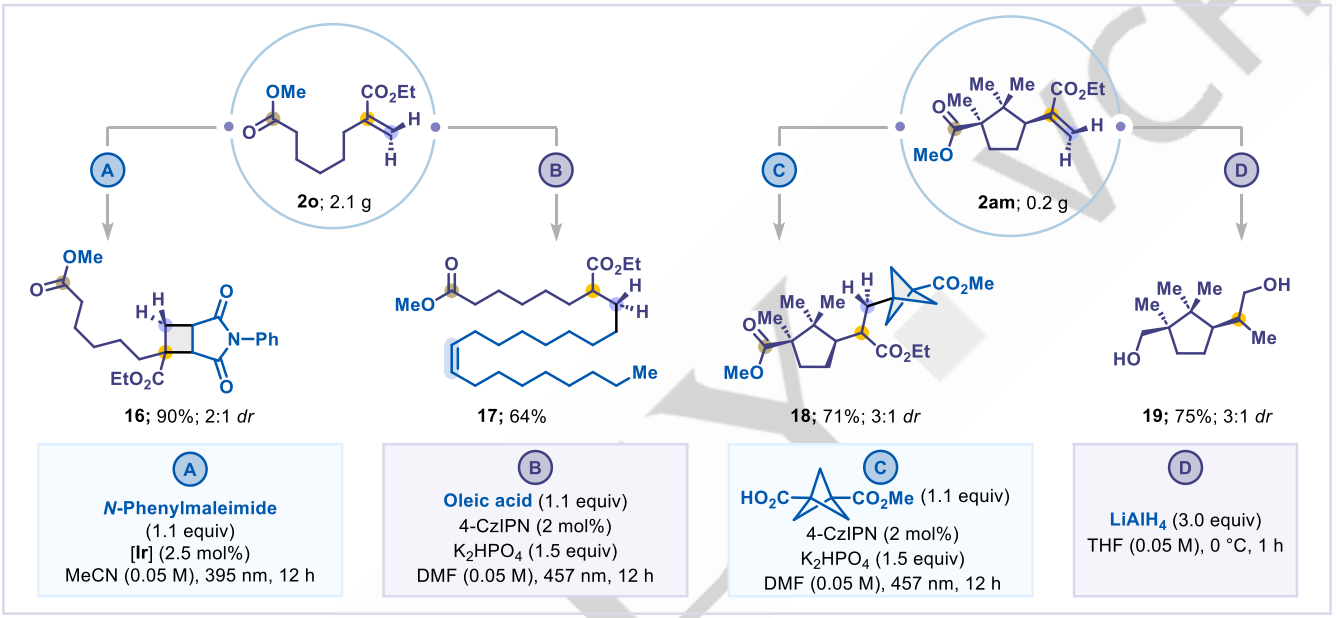


Figure 6: Post-synthetic modifications.

Mechanistic Insights

The possible reaction pathways for the electrochemical-interrupted Dowd-Beckwith reaction are illustrated in Figure 7A. Initially, deprotonation of **1a** generates carboxylate **I**, which undergoes single-electron transfer (SET) oxidation followed by decarboxylation to give the *primary* radical intermediate **II**. The deconstruction step could proceed *via* two distinct pathways: a) the radical intermediate **II** could undergo further anodic oxidation followed by fragmentation of **III** to give the acylium ion **IV**,[22] or b) intermediate **II** may undergo *β*-scission to give acyl radical **V** followed by anodic oxidation to acylium ion **IV**. Finally, methanol is added to the acylium ion **IV** to furnish product **2a**. To elucidate the favorable pathway, we conducted a series of mechanistic experiments to probe the involvement of radical, *primary* carbocation, and acylium ion intermediates. Electrolysis of **1a** under standard conditions in the presence of radical scavengers, such as TEMPO or BHT, completely inhibits the formation of **2a**, indicating the involvement of a radical intermediate **II** (Figure 7B).

Next, electrolysis of carboxylic acids **1ak** and **1al** under standard conditions resulted in the formation of methanol addition products (**20** and **22**) and olefins **(21** and **23)**, suggesting the involvement of *primary* carbocation intermediate (Figure 7C, experiments 1 and 2). In addition, electrolysis of **1a** in the absence of methanol, followed by treatment of the reaction mixture after electrolysis with methanol, afforded the deconstructive product **2a** in 20% yield (Figure 7C, experiment 3), supporting the formation of the acylium ion **IV**. DFT calculations (Figure 7D) indicate that the radical ring-expansion pathway is more favorable than the fragmentation pathway,[23] suggesting that the competing pathways are ring-expansion vs anodic oxidation, not radical fragmentation. However, we didn't observe the formation of ring-expansion product **3**. Overall, the combined mechanistic experiments and DFT studies support that the radical intermediate **II** undergoes fast anodic oxidation to give *primary* carbocation **III** that would fragment to afford acylium ion **IV**,[22] which then reacts with methanol to deliver the final product **2a**.

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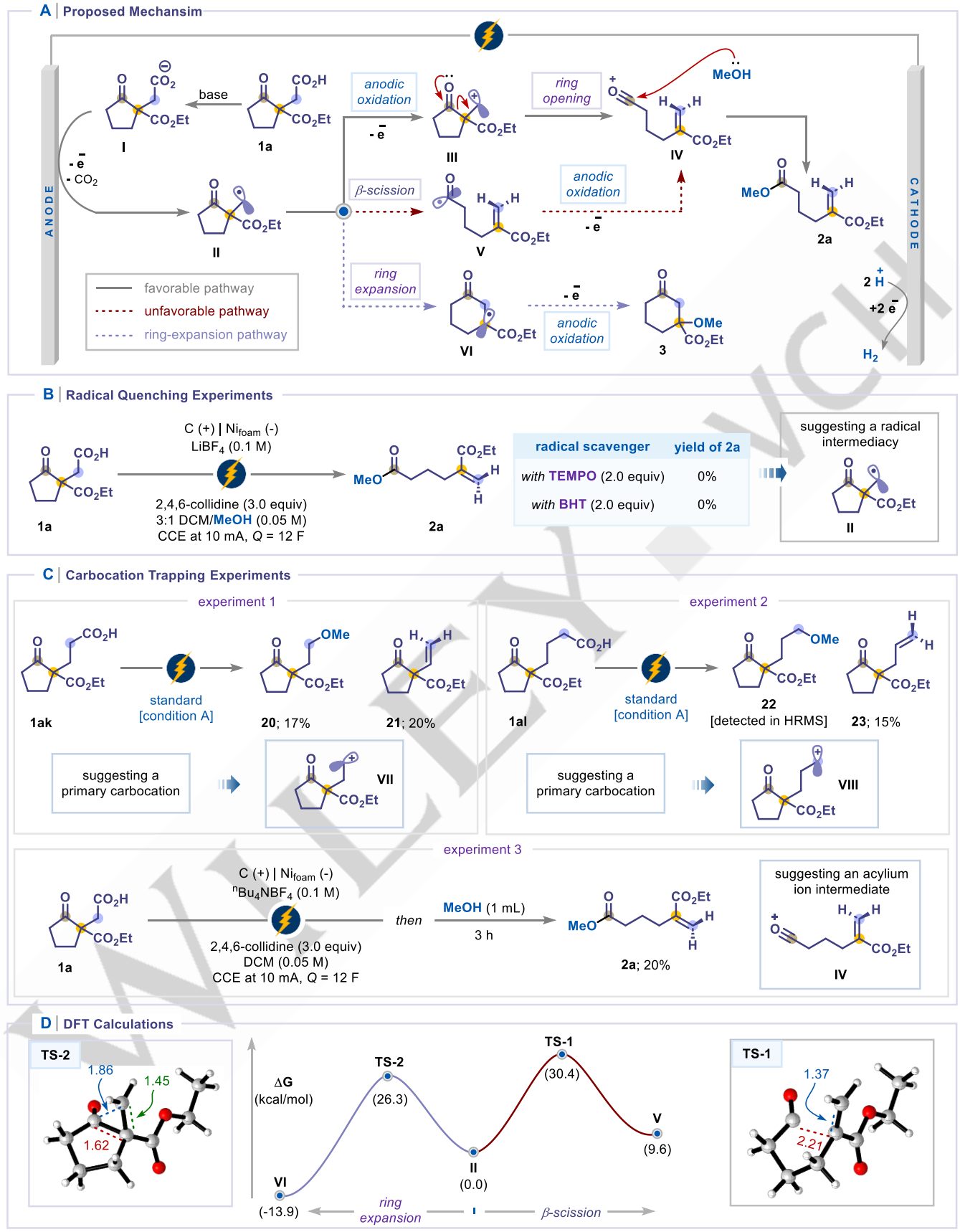


Figure 7: (A) Proposed mechanism. (B) Radical quenching experiments. (C) Carbocation trapping experiments. (D) DFT calculations. Calculations were performed at ROMP2/6-311++G(d,p)//UB3LYP/6-311++G(d,p) level of theory. Bond distances are in Å unit. Relative Gibbs free energies are in kcal/mol.

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Conclusion

In summary, we have developed the first regioselective deconstructive functionalization of cyclic ketones *via* an interrupted Dowd-Beckwith reaction under electrochemical conditions. This protocol demonstrates broad applicability across a range of cyclic ketones. The reaction is practical, scalable, and tolerates various sensitive functional groups. A broad range of naturally occurring complex ketones were successfully deconstructed into intricate acyclic molecules, which are challenging to achieve using traditional methods. The synthetic potential of the developed strategy was demonstrated in the asymmetric synthesis of dimethylcyclobutane-containing natural products, planococcol, maconelliol, and their ester derivatives. Furthermore, the functional groups introduced during the transformation served as valuable synthetic handles for accessing important building blocks. Mechanistic experiments suggest the involvement of *primary* carbocation followed by deconstructive functionalization. Given the novel reactivity and the increasing demand for deconstructive strategies in scaffold hopping within drug discovery programs in medicinal chemistry, we foresee that our strategy will have valuable applications in both synthetic and medicinal chemistry.

Supporting Information

General information, experimental procedures, characterization data for all new compounds, DFT calculations, and NMR spectra are available in the supplementary information.

Acknowledgements

Financial support from the Science and Engineering Research Board (SERB), Government of India (File CRG/2022/007372) is gratefully acknowledged. D. P. H. thanks the Indian Institute of Science (IISc) Bangalore for the start-up grant and infrastructure. T. S. thanks the Ministry of Education, Government of India, for PMRF. J. V. N. K. thanks the University Grant Commission (U.G.C.) for doctoral fellowship.

Keywords: Deconstructive functionalization • Dowd-Beckwith reaction • Electrochemistry • Cyclic ketones • Total synthesis

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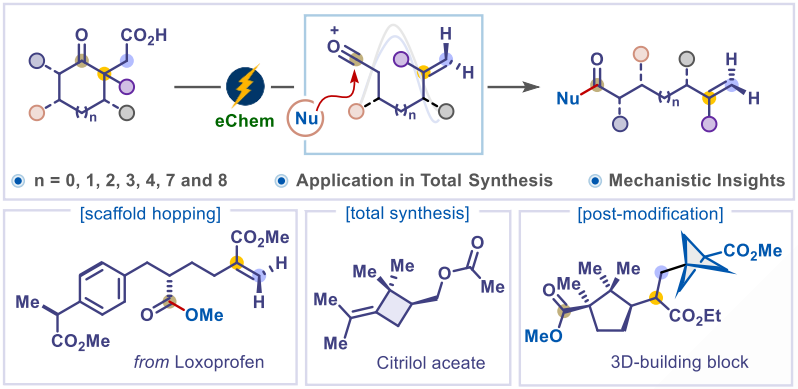
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10.1002/anie.202505155

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RESEARCH ARTICLE

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Interrupted Dowd-Beckwith (IDB) reaction enables the highly regioselective deconstructive functionalization of cyclic ketones using electricity as the sole oxidant. This strategy transforms complex cyclic ketones into acyclic molecules and has been applied to the asymmetric synthesis of planococcol, citrilol acetate, maconelliol and its derivatives. Mechanistic studies support an oxidative radical-polar crossover.

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